

Exercise 1

Consider the model from Exercise 1 on Sheet 4:

$$\begin{aligned} \text{wage} = & \beta_0 + \beta_1 \cdot \text{age} + \beta_2 \cdot \text{HSGrad} + \beta_3 \cdot \text{SomeCollege} + \beta_4 \cdot \text{CollegeGrad} \\ & + \beta_5 \cdot \text{AdvancedDegree} + \beta_6 \cdot \text{health.Verygood} + \varepsilon. \end{aligned}$$

with

$$\hat{\beta}'_{LS} = (52.61, 0.62, 11.01, 23.16, 37.97, 62.63, 9.13)$$

Hint: To calculate the following solutions you need Sheet 3 and 4.

- (a) Calculate the corrected coefficient of determination.
- (b) Assume Gaussian errors and calculate the AIC and the BIC of the model. Please note that the book uses $\log(x)$ as the natural logarithm with base e (and not base 10)!

Exercise 2

Now consider the model with all available variables. The R output of the model is as follows

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	-2.423e+03	6.165e+02	-3.931	8.67e-05	***
year	1.241e+00	3.074e-01	4.037	5.54e-05	***
age	2.707e-01	6.223e-02	4.350	1.41e-05	***
maritl2. Married	1.718e+01	1.720e+00	9.985	< 2e-16	***
maritl3. Widowed	2.052e+00	8.005e+00	0.256	0.79774	
maritl4. Divorced	3.967e+00	2.887e+00	1.374	0.16951	
maritl5. Separated	1.153e+01	4.844e+00	2.380	0.01736	*
race2. Black	-5.096e+00	2.146e+00	-2.375	0.01760	*
race3. Asian	-2.814e+00	2.603e+00	-1.081	0.27978	
race4. Other	-6.059e+00	5.666e+00	-1.069	0.28505	
education2. HS Grad	7.759e+00	2.369e+00	3.275	0.00107	**
education3. Some College	1.834e+01	2.520e+00	7.278	4.32e-13	***
education4. College Grad	3.124e+01	2.548e+00	12.259	< 2e-16	***
education5. Advanced Degree	5.395e+01	2.811e+00	19.190	< 2e-16	***
jobclass2. Information	3.571e+00	1.324e+00	2.697	0.00704	**
health2. >=Very Good	6.515e+00	1.421e+00	4.585	4.72e-06	***
health_ins2. No	-1.751e+01	1.403e+00	-12.479	< 2e-16	***

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Residual standard error: 34 on 2983 degrees of freedom

Multiple R-squared: 0.3396, Adjusted R-squared: 0.3361

F-statistic: 95.89 on 16 and 2983 DF, p-value: < 2.2e-16

- (a) Interpret the effect of one continuous, one binary and one categorical covariate of your choice.
- (b) Which effects are significantly different from 0 at the significance level $\alpha = 0.05$?
- (c) Calculate the AIC and the BIC of the model. Use $\hat{\epsilon}'\hat{\epsilon} = 3448498$.
- (d) Compare this model with the model from Exercise 1. Take the corrected coefficient of determination, the AIC and the BIC into account. Which model do you prefer? Justify your answer!